



Cambrian College

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality

Course Outline

Course Title	Requirements Elicitation				
Course Code:	BTA1003	Credit Value:	4	Credit Hours:	56
Programs:	BAGC I.T. Business Analysis				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

General Education Course: ☐

Eligible for PLAR: ☒

COURSE DESCRIPTION

In this course, students will learn techniques for developing and managing valid and applicable requirements to a wide range of software projects. Learners will adapt and scale these practices best to meet the needs of a particular business situation. Various tools are introduced and used to facilitate communication and help future software practitioners, managers, marketers and customers apply practical requirements development methods. The techniques covered in this course are suitable for many types of projects of any size including both new development and enhancements. Additionally, these skills will benefit students when working with both local and distributed teams using either traditional or agile development methods.

Date: June 19, 2020

Prepared By: Nathan Abourbih

Approved by: 

BRIAN LOBBAN
Dean, School Of Business And Information Technology,
Creative Arts, Design, Music And Hospitality

Effective: Fall 2020, Winter 2021, Spring 2021

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL	
This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:	
Program(s)	Vocational Learning Outcomes
I.T. Business Analysis	College Standards 1. Plan and manage the requirements gathering activities throughout the requirements process for changes to a business system 2. Analyze the information collected during the requirements elicitation to identify gaps and define the capabilities of an acceptable solution using appropriate methods, tools, techniques and documentation. 3. Document business requirements in accordance with standard business practices. 4. Communicate the results of business requirements analysis and documentation to a broad and diverse audience using appropriate communication tools and styles. 5. Apply negotiation and problem-solving skills to build effective stakeholder alignment.

COURSE CURRICULUM

Topics/Concepts Covered in This Course

- Software Requirements: What, Why, and Who
- Requirements Development
- Requirements for Specific Project Classes
- Tools for Requirements Development

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	Objectives
1. Distinguish key terms used in the software requirements domain.	1.1 Define and describe: 1.1.1 Product requirements from project requirements. 1.1.2 Requirements development from requirements management. 1.1.3 Several requirements-related problems. 1.2 Discuss the consequences of insufficient user involvement, inaccurate planning, creeping user requirements, ambiguous requirements, gold plating and overlooked stakeholders. 1.3 Identify the benefits of a high-quality requirements process.

Learning Outcome	Objectives
2. Describe the customer-development relationship critical to software project success.	2.1 Refer to and consistently use the Requirements Bill of Rights and the Requirements Bill of Responsibilities for software customers. 2.2 Discuss the critical issue of reaching agreement on a set of requirements planned for a specific release or development iteration: 2.2.1 Create a culture that respects requirements. 2.2.2 Identify decision-makers. 2.2.3 Reach agreement on requirements to produce a baseline. 2.2.4 Deal with analysis paralysis.
3. Use the best-practice approach and stock a software tool kit with a variety of techniques that can be applied to diverse problems.	3.1 Describe and follow a requirement development process framework (elicitation, analysis, specification, validation, requirements management, knowledge, project management) as an iterative process. 3.2 Select, apply and adapt a set of practices most appropriate for understanding the requirements for different portions of a given project in its lifecycle.
4. Describe the vital functions the business analyst performs, the skills and knowledge an effective analyst needs, and how to develop BAs in an organization.	4.1 Distinguish the business analyst role, job titles, and associated team members in today's organizations. 4.2 Describe typical activities performed by the business analyst. 4.3 Practice and demonstrate essential analyst soft skills: listening, interviewing and questioning, critical thinking, analytical thinking, system thinking, learning, facilitation, leadership, observational, communication, organizational, modeling, interpersonal, and creative skills. 4.4 Demonstrate a breadth of business analyst knowledge: 4.4.1 Apply contemporary requirements engineering practices and apply these in the context of various software development life cycles, risk management, and quality engineering. 4.4.2 Describe the architecture and operating environment. 4.4.3 Apply business, industry and the organization concepts. 4.5 Describe the competencies that entry-level, junior, intermediate, and senior business analysts should exhibit according to International Institute of Business Analysis (IIBA). 4.6 Explore how people with different backgrounds move into the analyst role and see some of the challenges and risks they will face.
5. Establish business requirements.	5.1 Identify the desired business benefits and resolve conflicting requirements. 5.2 Write vision and scope document deliverables.

Learning Outcome	Objectives
	5.3 Explore several scope representation techniques and models that can be included in scope documents: context diagram, an ecosystem map, feature tree, event list. 5.4 Control the scope with established business objectives and by assessing the impact of changes. 5.5 Manage scope on an agile project. 5.6 Determine project completion using business objectives.
6. Use techniques to engage classes of users in software development.	6.1 Identify and establish user classes. 6.2 Document the user classes and their characteristics, responsibilities, and physical locations in software requirements specification (SRS) or in a requirements plan for the project. 6.3 Establish user personas as representatives of their user class, based on market, demographic, and ethnographic research. 6.4 Explore the product champion approach to structure customer-development collaborative partnerships. 6.5 Describe user representation on agile projects.
7. Describe a variety of effective elicitation techniques, the overall elicitation process, as well as cautions and suggestions for adopting this process.	7.1 Enumerate and follow the activities of elicitation: prepare, perform and follow up after elicitation. 7.2 Use either or, in combination, interviews, workshops, focus groups, observations, questionnaires, system interface analysis, user interface analysis, document analysis, to discover information regarding project requirements. 7.3 Plan elicitation on a project. 7.4 Classify customer input to produce a list of needs: 7.4.1 Business Requirements 7.4.2 User Requirements 7.4.3 Business Rules 7.4.4 Functional Requirements 7.4.5 Quality Attributes 7.4.6 External Interface Requirements 7.4.7 Constraints 7.4.8 Data Requirements 7.4.9 Solution Ideas 7.5 Determine the point of diminishing returns on elicitation requirements. 7.6 Identify techniques and approaches to minimize assumed, implied or missing requirements. 7.7 Discuss the use of user-centric and usage-centric approaches to identify and prioritize necessary requirements.

Learning Outcome	Objectives
8. Discover and define business rules that influence business processes by establishing vocabulary, imposing restrictions, triggering actions, and governing how computations are carried out.	8.1 Distinguish between business rules, processes and requirements. 8.2 Use the business rules taxonomy from the perspective of both business and its information systems to classify and document rules in a consistent way as: 8.2.1 Facts 8.2.2 Constraints 8.2.3 Action Enablers 8.2.4 Inferences 8.2.5 Computations 8.2.6 Atomic Business Rules 8.3 Document business rules as enterprise-level assets, selecting appropriate tools for the organization type. 8.4 Describe ways and processes for discovering business rules. 8.5 Determine which rules require implementation in the software.
9. Document the software requirements specification (SRS), purpose, structure and contents, which is delivered to designers, builders and verifiers of the solutions.	9.1 Define the SRS as standardized by ISO/IEC/IEEE2011 and identify the audiences who rely on it. 9.2 Baseline an agreement for each set of requirements before team implementation. 9.3 Select and adhere to unique and persistent numbering or textual tag labelling format in the documentation. 9.4 Balance conceptual images and screen layouts in documenting interfaces in SRSs. 9.5 Use and adapt templates for SRS with effective version control labels. 9.6 Contrast requirements specification on agile projects with respect to: 9.6.1 Stories documentation. 9.6.2 Stories allocated to iterations. 9.6.3 User acceptance tests outlining exception conditions and expected behaviour.
10. Write desirable characteristics of requirements and of requirements documents.	10.1 Verify that requirements and requirements collections exhibit complete, correct, feasible, necessary, prioritized, unambiguous, verifiable, consistent, modifiable and traceable qualities.

Learning Outcome	Objectives
	10.2 Adopt industry-standard guidelines for writing requirements: 10.2.1 Write from the system and user perspective. 10.2.2 Write the statement of need or functionality first, followed by supporting details. 10.2.3 Incorporate an appropriate and consistent level of detail. 10.2.4 Enhance natural language requirements with lists, tables, visual analysis models, charts, mathematical formulas, photographs, sound clips, and video clips. 10.2.5 Avoid ambiguous terms that lead to unverifiable requirements. 10.2.6 Avoid incompleteness. 10.3 Observe problems with requirements and suggest improvements.
11. Demonstrate the use of diagrams and industry modeling techniques to create visual representations of the requirements.	11.1 Discuss the various types of diagrams and visual models used to represent requirements. 11.2 Demonstrate the ability to read, interpret, select and prepare the correct visual models for distinct types of information requirements. 11.3 Explain the importance of relating customer narrations and descriptions to analysis model components.
12. Specify and document data requirements from various models.	12.1 Discuss methods of modeling data relationships in order to analyze the problem domain. 12.2 Describe analysis techniques used to rule out gaps, errors and inconsistencies in requirements. 12.3 Populate data dictionary entries to house information about composition, data types, allowed values into a shared resource. 12.4 Use a template approach to help in soliciting distinct aspects of reports and information to record. 12.5 Specify the requirements for a dashboard report to provide a consolidated, multidimensional view of an organization or a process.
13. Distinguish requirements gathering techniques between classes of projects.	13.1 Describe the characteristics of agile approaches as they relate to the requirements for an agile project. 13.2 Provide specific suggestions as to which practices are most relevant and how to use them in enhancement or replacement of existing information systems projects. 13.3 Define and evaluate requirements for packaged solution projects such as commercial off the shelf (COTS) products or cloud solutions. 13.4 Describe the elements and considerations involved in a variety of different project classes.

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.
- respond to written, spoken, or visual messages in a manner that ensures effective communication.

Numeracy

- execute mathematical operations accurately.

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.
- use a variety of thinking skills to anticipate and solve problems.

Information Management

- locate, select, organize, and document information using appropriate technology and info systems.
- analyze, evaluate, and apply relevant information from a variety of sources.

Interpersonal

- show respect for the diverse opinions, values, belief systems, and contributions of others.
- interact with others in groups in ways that contribute to effective working relationships.

Personal

- not applicable

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Online: Course is fully delivered through asynchronous teaching.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.
- HyFlex: Course includes both synchronous and asynchronous learning and the student can move between both components seamlessly.

Learning Activities

- Lectures
- Class Discussions
- Group Work
- Research
- Presentations
- Guest Speaker(s)
- In-Class Exercises
- Case Studies

Resources Required

Books

Karl E Wieggers, Joy Beatty, *Software Requirements 3, 3rd Edition*, 3rd, Microsoft Press
ISBN: 0-7356-7966-5

Evaluation Plan

Grading Scheme

A	80% - 100%
B	70% - 79%
C	60% - 69%
D	50% - 59%
F	0% - 49%

Evaluation Method	Value (%)
Assignments	40%
Applied Activities(G)	60%
Case Studies (30%) Presentation(s)/Simulation(s) (30%)	

ADDITIONAL INFORMATION

College

Academic Policies

Students must adhere to all academic policies. These are available on myCambrian as well as the College website at <https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/>.

Prior Learning Assessment and Recognition (PLAR)

Students may contact the Pathways office at pathways@cambriancollege.ca

Transfer Credit

Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Accessibility Services

Cambrian is committed to creating an inclusive learning environment where we support our students' success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@cambriancollege.ca

Testing Practices at Cambrian

Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Wabnode Centre for Indigenous Services

Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling or academic tutoring. Our team will also be able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode for cultural, sports and community activities. Students can also contact Wabnode at wabnode@cambriancollege.ca

Copyright

Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: <http://cambriancollege.libguides.com/copyrightinfo> or by contacting the Library for questions related to copyright: library@cambriancollege.ca

Out-of-Class Assistance

The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Audio/Visual Capture

Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Equity, Diversity and Inclusivity

Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: <https://cambriancollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/>

Tests and Evaluations

Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Attendance

Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Modifications to the Course Evaluation Plan

Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).